Life Sciences and
Facility Management
Institute of
Computational Life Sciences

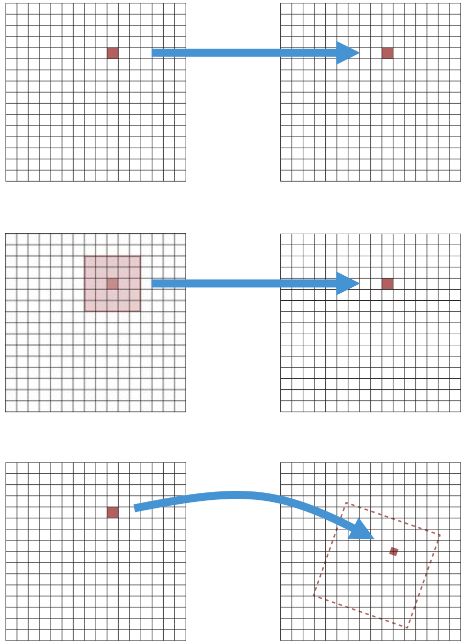
Spectral filtering

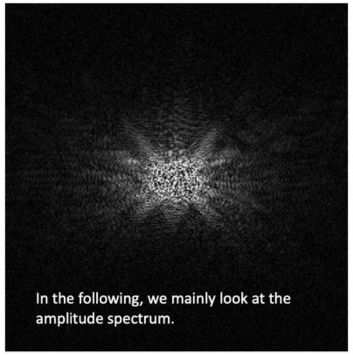
Image and Signal Processing

Norman Juchler

Recap of last week's topics

- Different types of image transformations
- 2D discrete Fourier transform





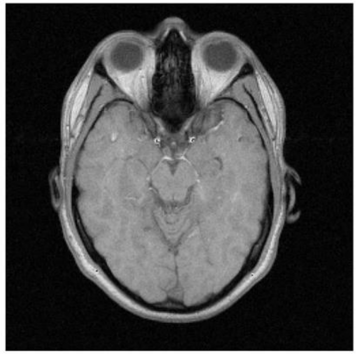


Illustration: Link

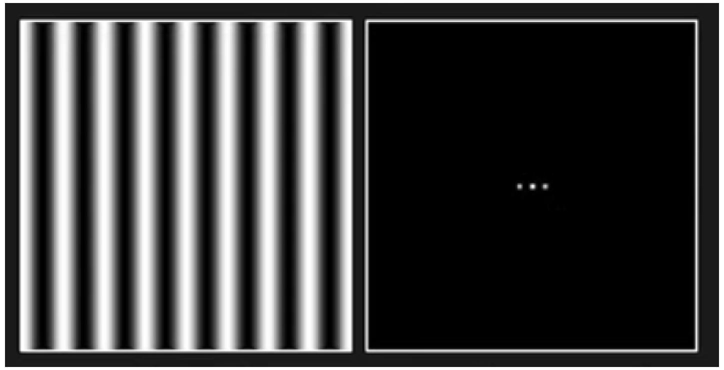
Sinusoidal gratings and Fourier transform

Examples:

- $f(x, y)$ and $g(x, y)$ are cos functions in the x-direction
- We look at the logarithm of the amplitude spectrum:

$f(x,y)$

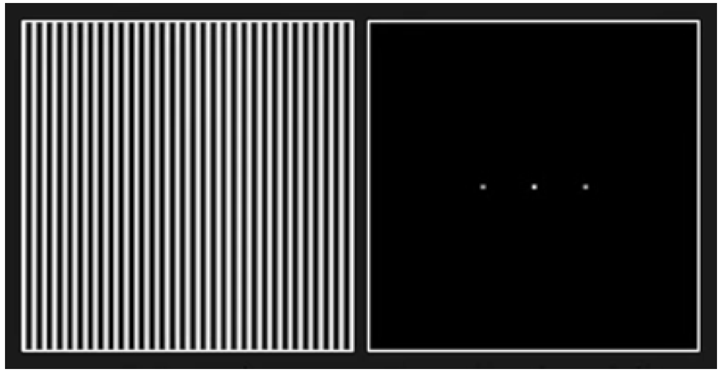
Periode =8



$\log(|F(u,v)|)$

$g(x,y)$

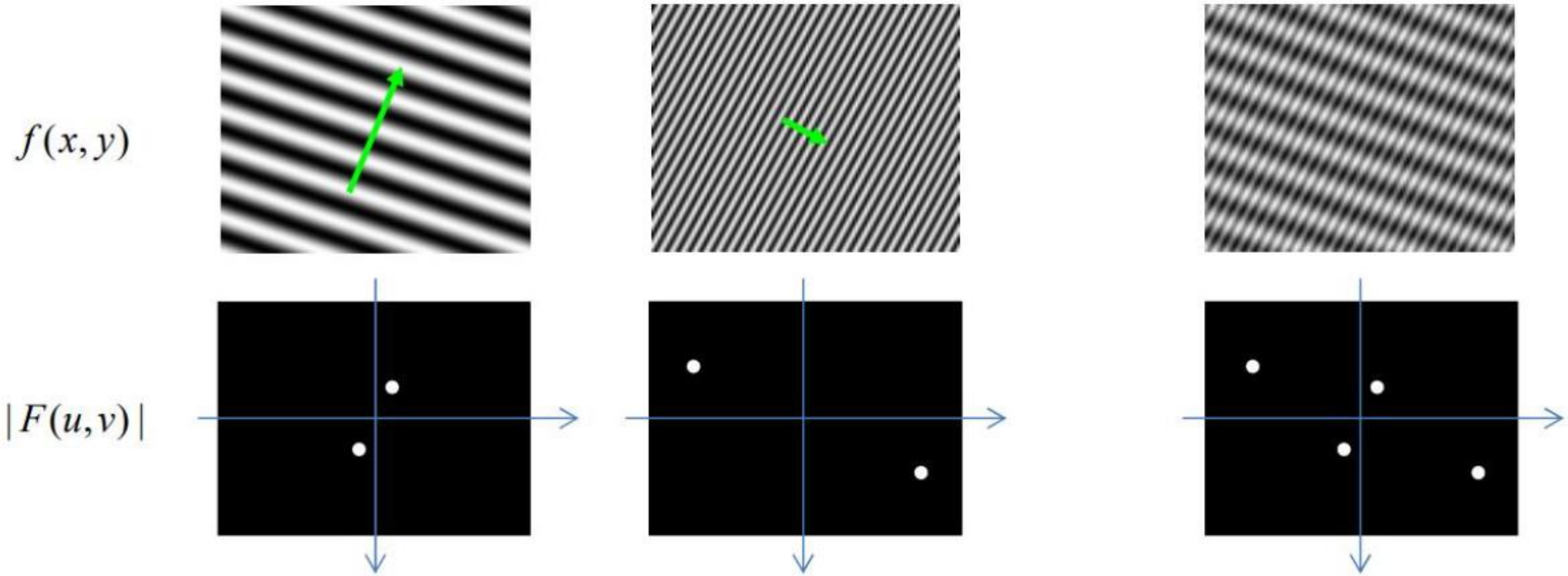
Periode =32



$\log(|G(u,v)|)$

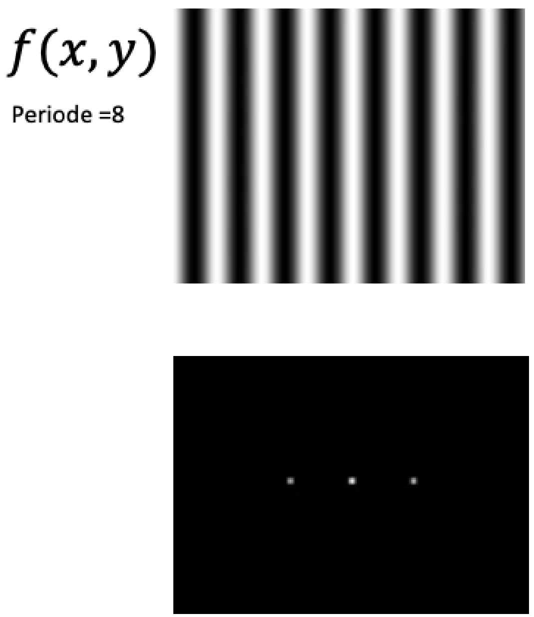
Sinusoidal gratings and Fourier transform

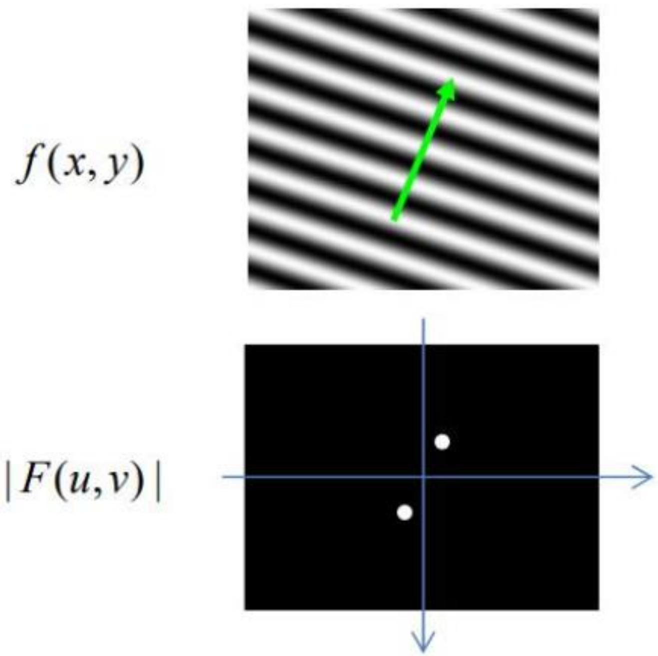
A wave in the image f(x,y) is always linked to a point pair in the amplitude spectrum |F(u,v)|. Thus, the distribution of the significant frequencies in the image can be recognized in the amplitude spectrum.



Wait a second! Is there an inconsistency?

- On the right, the DC component is 0, whereas on the left, the DC component is > 0. Why?
- Resolution:
  - The input image on the left was an 8-bit integer image (with mean value > 0)
  - The input image on the left was a floating point image (with mean value = 0)





Where has the dot in the middle gone?

2D Discrete Fourier transform

continued

2D Discrete Fourier Transform: 2D DFT

- Input: Image $f(x,y)$ of size $M \times N$
- The 2D DFT is defined as

$$\boxed{F(u,v) = \frac{1}{MN} \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f(x,y) e^{-i2\pi\left(\frac{ux}{M}+\frac{vy}{N}\right)}}$$

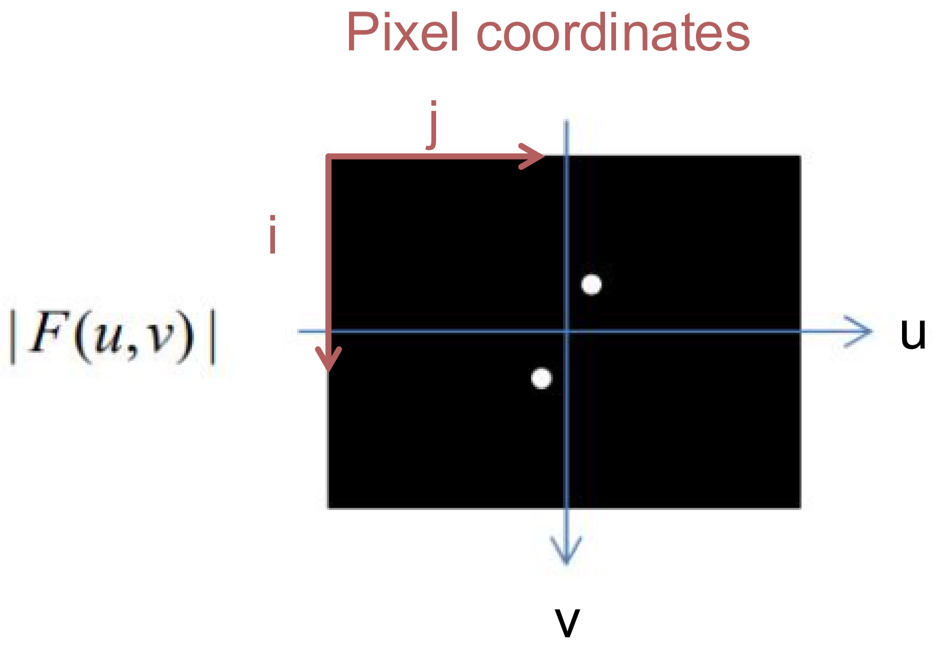
- The 2D inverse DFT (2D IDFT) is defined as

$$\boxed{f(x,y) = \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} F(u,v) e^{i2\pi\left(\frac{ux}{M}+\frac{vy}{N}\right)}}$$

- Summation formulas are separable (first x, then y), and symmetric (x then y, or y then x)
- Therefore, we can use 1D DFT to compute 2D DFT (good news for FFT)

DFT spectra as images

- Recall:
  - $F(u,v)$ in general is complex
  - $|F(u,v)|$ refers to the amplitude, and $\angle(F(u,v))$ to the phase
- We can represent amplitude (or phase) spectra of DFTs as images (`img = `$|F(u,v)|$)
- The spectral images have the same dimensions as the input images!
- **Note**: The spectra include negative frequencies! This means that: $|F(0,0)|$ = `img[h//2, w//2]`

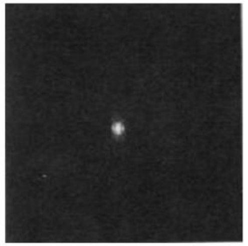


Visualization of the amplitude spectrum on a logarithmic scale

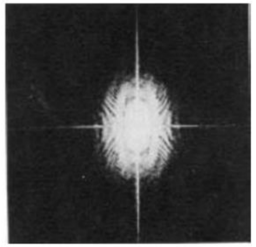
- The range of values of $F(u,v)$ is typically very large.
- Due to quantization of $F(u,v)$ small values are not distinguishable when we attempt to display the amplitude of $F(u,v)$.
- We, therefore, apply a logarithmic transformation to enhance small values: $D(u,v) = c \cdot \log(1+|F(u,v)|)$
- The parameter c can be chosen so that the range of $D(u,v)$ is [0,255], i.e

$$c = \frac{255}{\log(1+\max\{|F(u,v)|\})}$$

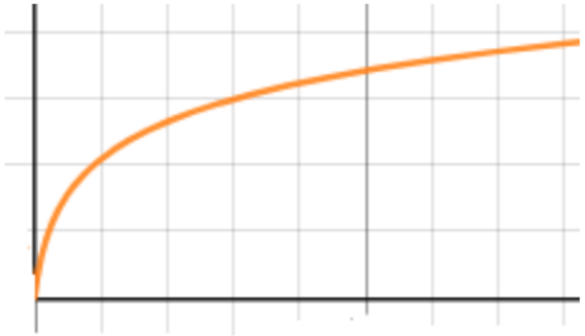
Original image



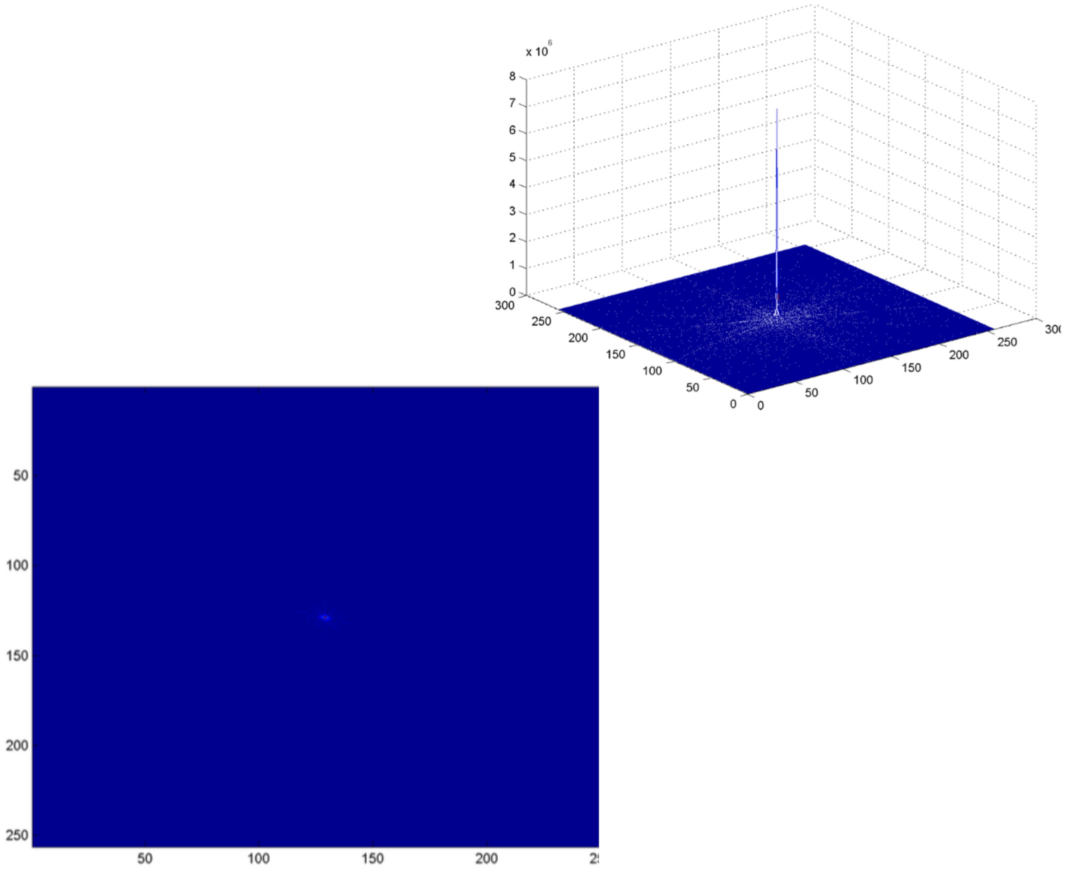
Display of amplitude of DFT



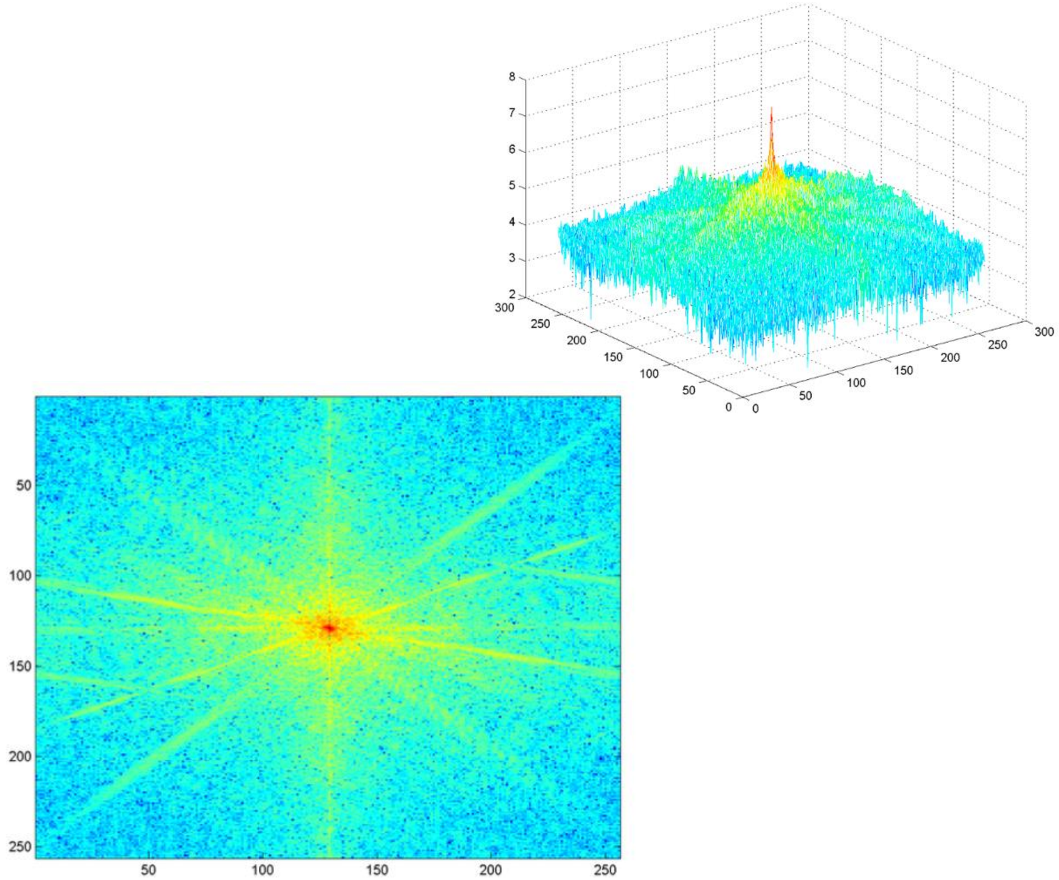
Display of the logarithmic amplitude of DFT



Visualization of the amplitude spectrum on a logarithmic scale



Display of the DFT amplitude



Display of the logarithmic DFT amplitude

Properties of the 2D (discrete) Fourier transform

Pretty much the same as we already know...

- **Periodicity**:
  - $F(u,v) = F(u+M,v) = F(u,v+N) = F(u+M,v+N)$
    ...with M, N being the width and height of the input image

- **Symmetries**:
  - If $f(x,y)$ is real and even then $F(u,v)$ is real and even.
  - If $f(x,y)$ is real and odd then $F(u,v)$ is imaginary and odd.

- **Linearity**:
  - $\mathcal{F}(f(x,y)+g(x,y)) = \mathcal{F}(f(x,y)) + \mathcal{F}(g(x,y))$,
    where $\mathcal{F}\{\cdot\}$ indicates the DFT operator.
  - This, however, does not imply that...
    $\mathcal{F}(f(x,y) \cdot g(x,y)) \neq \mathcal{F}(f(x,y)) \cdot \mathcal{F}(g(x,y))$

Properties of the 2D (discrete) Fourier transform

Pretty much the same as we already know...

- **Translation**: ("spatial delay")
  - Translation affects only the phase of the transform!

$$f(x - x_0, y - y_0) \leftrightarrow F(u, v)e^{-i2\pi\left(\frac{ux_0}{M} + \frac{vy_0}{N}\right)}$$

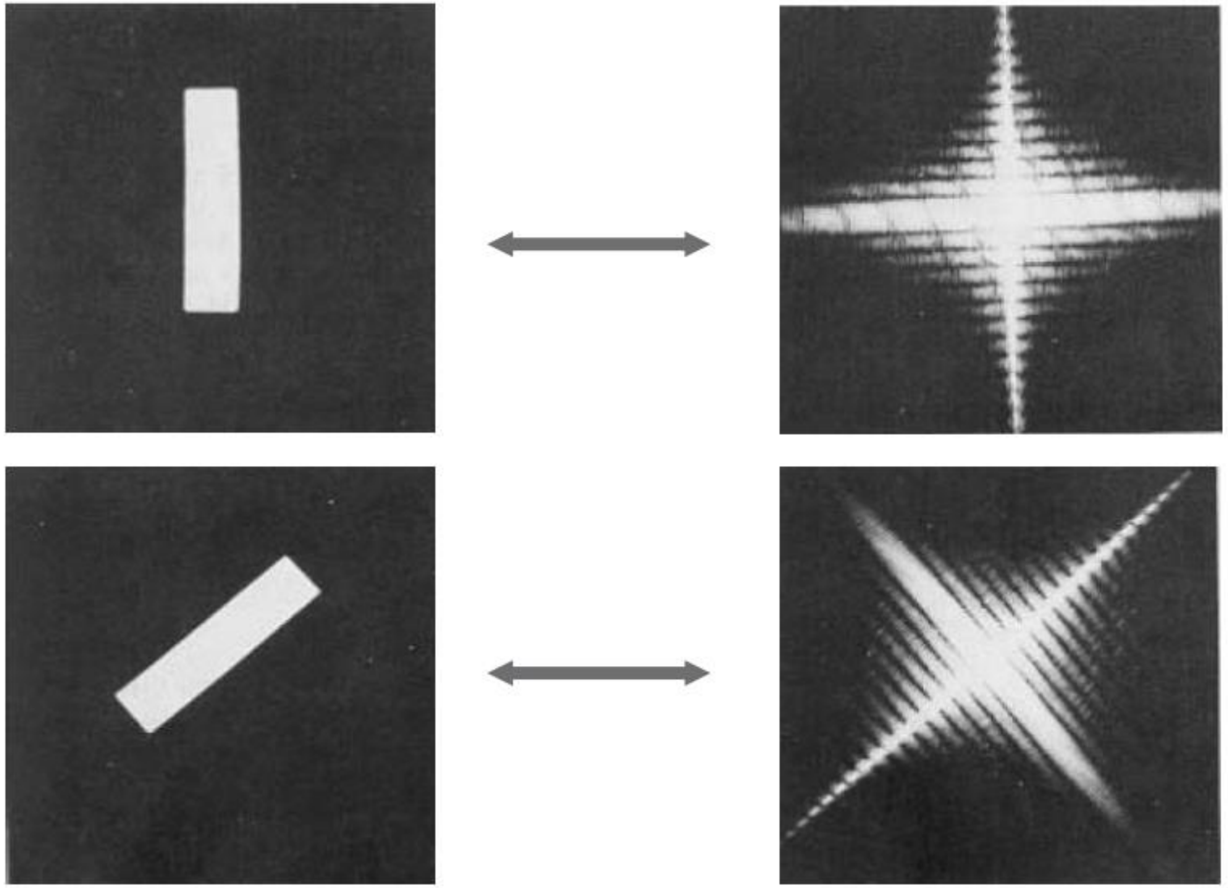
- **Average value**:
  - The DC component is equivalent to the mean signal $f(x,y)$

$$F(0,0) = \bar{f}(x,y) = \frac{1}{MN}\sum_x \sum_y f(x,y)$$

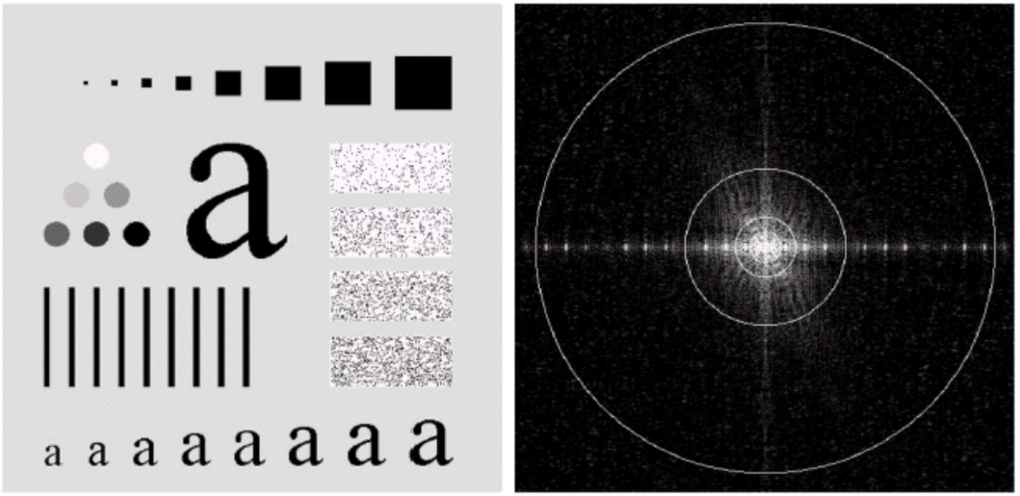
- **Convolution:**
  - $\mathcal{F}(f * g)(x,y) = \mathcal{F}(f(x,y)) \cdot \mathcal{F}(g(x,y))$,
    Convolution boils down to multiplication in the frequency domain

2D DFT and rotations

- Rotation of the image $f(x,y)$ by θ rotates $F(u,v)$ by θ

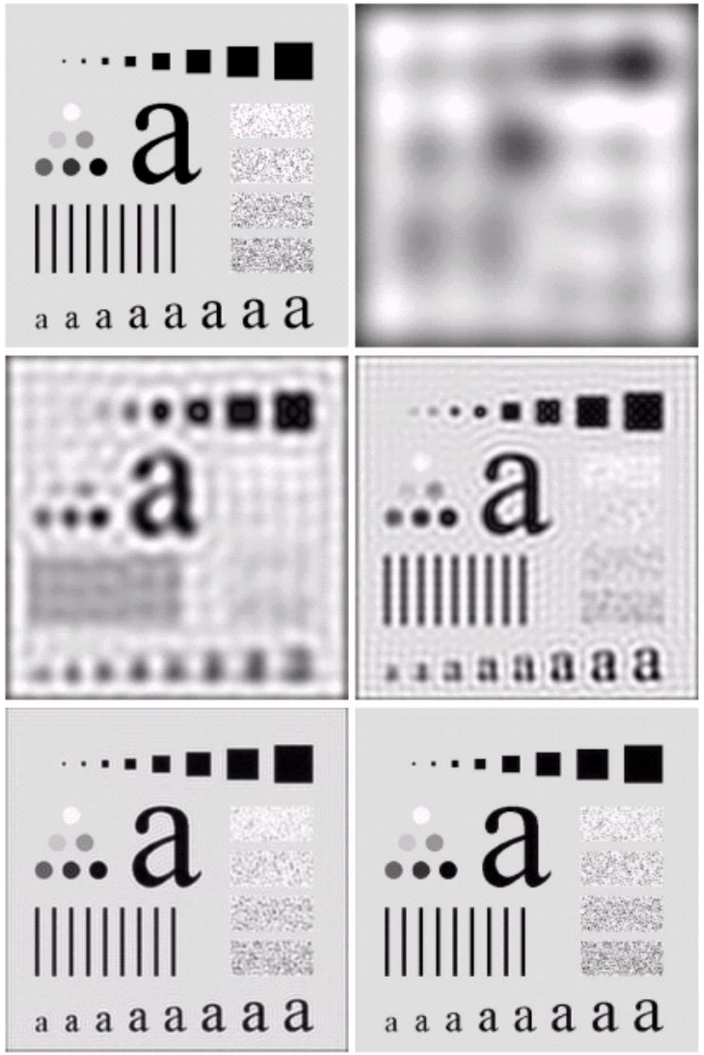


Applications: Low-pass filtering



a b

FIGURE 4.11 (a) An image of size 500 × 500 pixels and (b) its Fourier spectrum. The superimposed circles have radii values of 5, 15, 30, 80, and 230, which enclose 92.0, 94.6, 96.4, 98.0, and 99.5% of the image power, respectively.

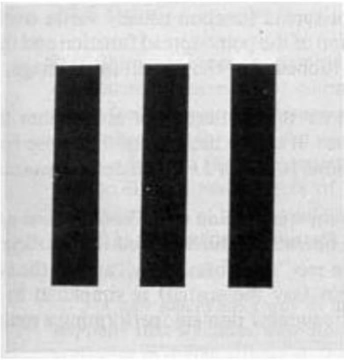


a b
c d
e f

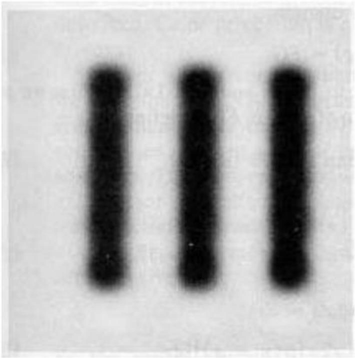
FIGURE 4.12 (a) Original image. (b)–(f) Results of ideal lowpass filtering with cutoff frequencies set at radii values of 5, 15, 30, 80, and 230, as shown in Fig. 4.11(b). The power removed by these filters was 8, 5.4, 3.6, 2, and 0.5% of the total, respectively.

Low- and high-pass filtering using amplitude of 2D DFT

- Low frequencies correspond to slowly varying pixel intensities (e.g., smooth surfaces)
- High frequencies correspond to quickly varying pixel intensities (e.g., edges)

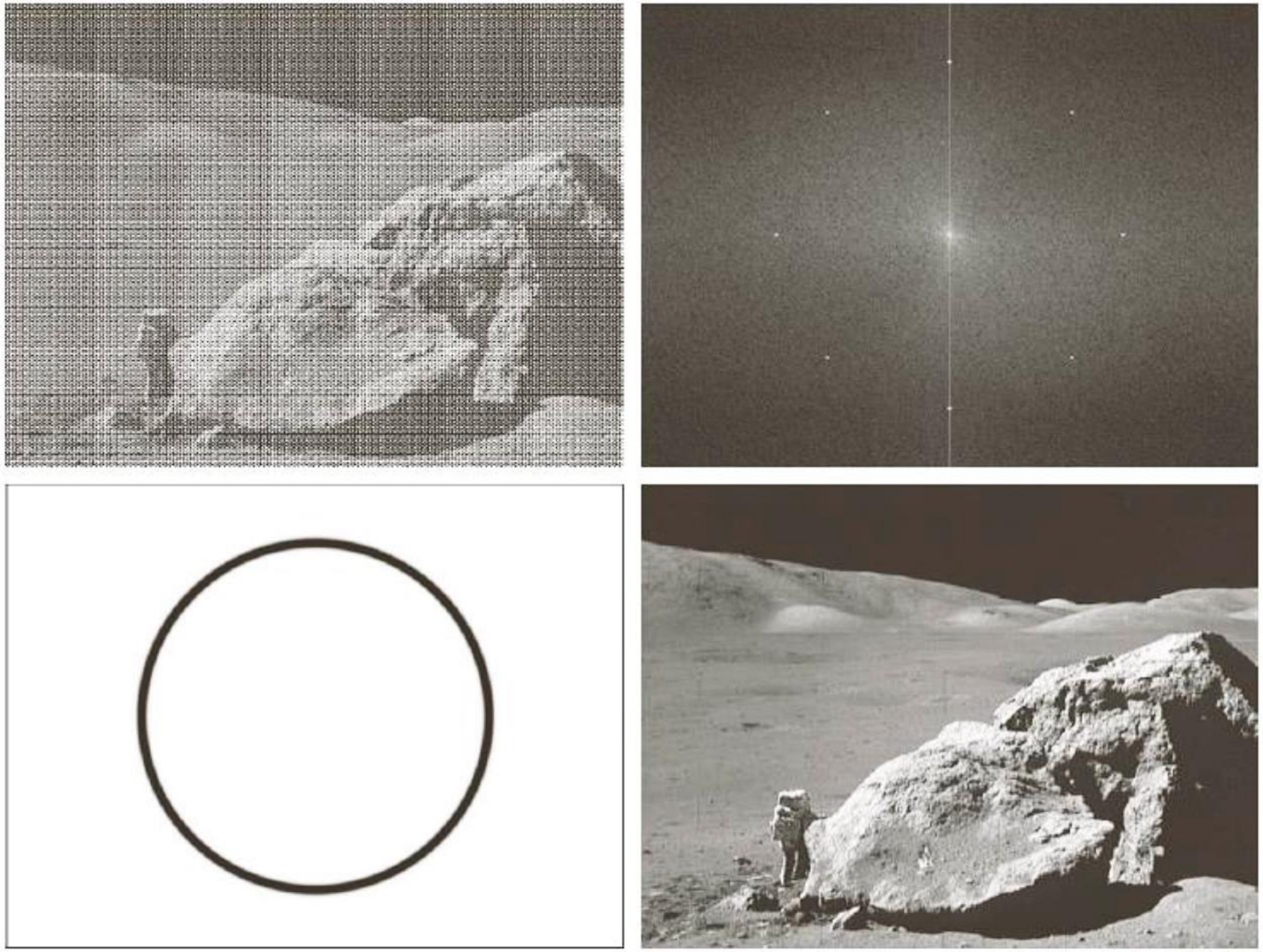


original image



low-pass filtered image

Low- and high-pass filtering using amplitude of 2D DFT



a b
c d

FIGURE 2.40
(a) Image corrupted by sinusoidal interference. (b) Magnitude of the Fourier transform showing the bursts of energy responsible for the interference. (c) Mask used to eliminate the energy bursts. (d) Result of computing the inverse of the modified Fourier transform. (Original image courtesy of NASA.)

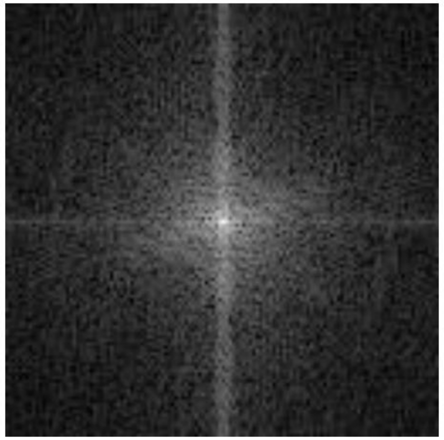
See Jupyter Notebook of the week!

Low- and high-pass filtering using amplitude of 2D DFT

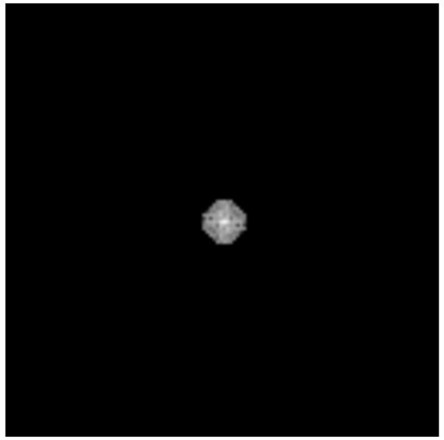
original image

low pass filtered

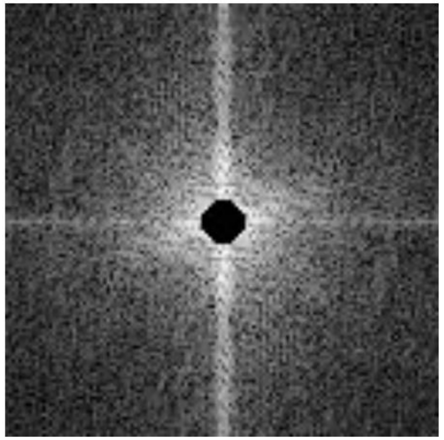
high pass filtered



amplitude of DFT



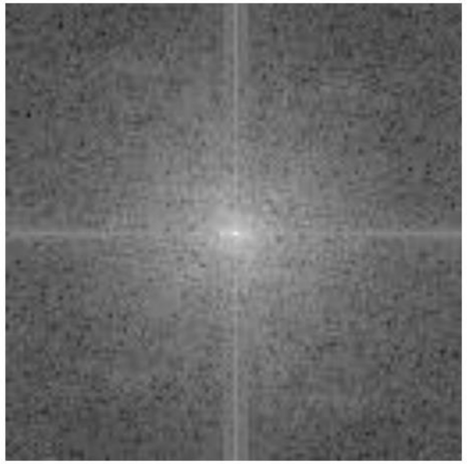
amplitude of DFT with
eliminated high frequencies

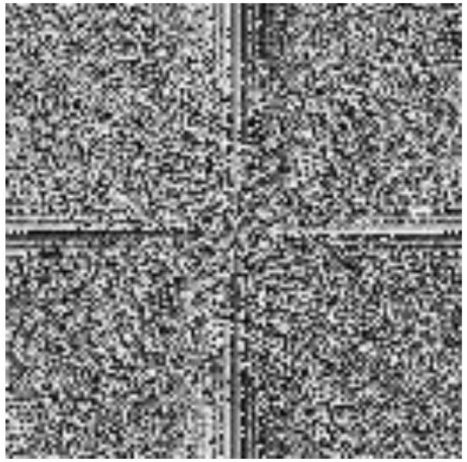


amplitude of DFT with
eliminated low frequencies

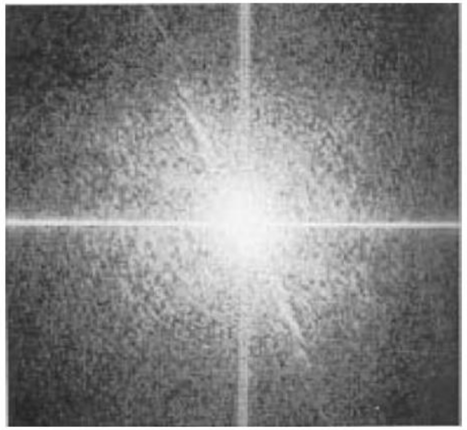
Amplitude vs. phase of 2D DFT

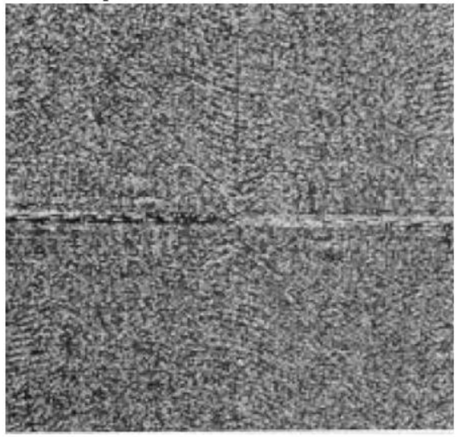

original image


amplitude of DFT

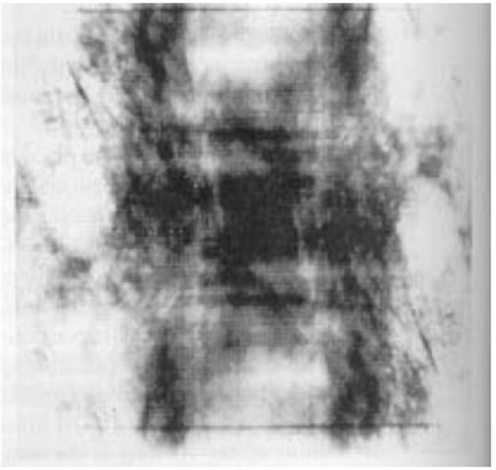

phase of DFT


amplitude of DFT


amplitude of DFT


phase of DFT

Amplitude vs. phase of 2D DFT

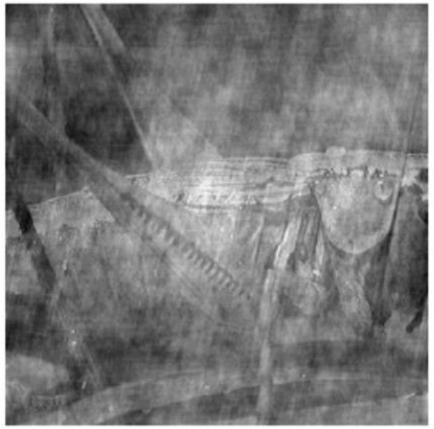


- Reconstructed image using the **amplitude of DFT only** and zero phase.
  - The magnitude determines the strength of each frequency component but not its location.

- Reconstructed image using the **phase of DFT only** and amplitude equal to 1.
  - The phase determines the locations of individual frequencies within the image.
  - High frequencies → abrupt changes → edges → object silhouettes → image content.

Amplitude vs. phase of 2D DFT

phase of cameraman +
amplitude of grasshopper



phase of grasshopper +
amplitude of cameraman